

Randomized Coordinate Descent under Polyak-Łojasiewicz Condition

1 Algorithm Description

The algorithm iteratively updates the full parameter vector by computing the gradient along a randomly selected coordinate direction at each step. It uses this coordinate-wise gradient information to determine the update direction and step size for the entire parameter vector. It is designed for functions satisfying the Polyak-Łojasiewicz condition that are smooth with respect to individual coordinates.

Assumptions

The algorithm assumes the objective function satisfies:

- **Polyak-Łojasiewicz (PL) Condition:** Ensures the gradient norm provides a lower bound on the suboptimality gap.
- **Coordinate-wise Smoothness:** Guarantees that the function's gradient is Lipschitz continuous along each coordinate.

Mathematical Formulation

Objective Function: Minimize a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, where f satisfies:

1. **Polyak-Łojasiewicz (PL) Condition:** There exists $\mu > 0$ such that:

$$\|\nabla f(w)\|^2 \geq 2\mu (f(w) - f^*), \quad \forall w \in \mathbb{R}^n,$$

where f^* is the minimum value of f .

2. **Coordinate-wise Smoothness:** For each coordinate $i \in \{1, 2, \dots, n\}$, the gradient $\nabla_i f(w)$ is Lipschitz continuous with constant L_i :

$$|\nabla_i f(w) - \nabla_i f(w')| \leq L_i |w_i - w'_i|, \quad \forall w, w' \in \mathbb{R}^n.$$

Update Rule: At iteration t , randomly select a coordinate $i_t \in \{1, 2, \dots, n\}$ uniformly at random. Update the parameter vector $w \in \mathbb{R}^n$ as:

$$w_{t+1} = w_t - \eta \cdot \nabla_{i_t} f(w_t) \cdot e_{i_t},$$

where:

- $e_{i_t} \in \mathbb{R}^n$ is the standard basis vector (1 at position i_t , 0 elsewhere).
- $\eta > 0$ is the step size, typically chosen as $\eta = \frac{1}{L_{i_t}}$ for coordinate-wise smoothness.

Hyperparameters:

1. **Step Size η :** Controls the magnitude of updates. For convergence under smoothness, $\eta \leq \frac{1}{L_{\max}}$, where $L_{\max} = \max_i L_i$.
2. **Maximum Iterations $T_{\max}$:** Limits computational effort.
3. **Tolerance ε :** Stops iterations when $\|\nabla f(w_t)\| < \varepsilon$.

2 Pseudo Code and Dry Run

Pseudocode

Input: Objective function f , step size η , max iterations $T_{\max}$, tolerance ε .

- 1: **Initialize:** $w_0 \in \mathbb{R}^n$.
- 2: **for** $t \in 1, 2, \dots, T_{\max}$ **do**
- 3: Randomly select coordinate $i_t \in \{1, 2, \dots, n\}$ uniformly.
- 4: Compute gradient partial derivative: $g_t = \nabla_{i_t} f(w_t)$.
- 5: Update parameter: $w_{t+1} = w_t - \eta \cdot g_t \cdot e_{i_t}$.
- 6: **if** $\|\nabla f(w_{t+1})\| < \varepsilon$: **break** (convergence check).
- 7: **end for**

Output: Final parameter vector w_{t+1} .

Dry Run Example

Function: $f(w) = (w - 3)^2$, with $\nabla f(w) = 2(w - 3)$. **Initialization:** $w_0 = 0$, $\eta = 0.1$, $\varepsilon = 0.01$, $T_{\max} = 3$.

Iteration 1:

- Gradient: $\nabla f(w_0) = 2(0 - 3) = -6$.
- Update: $w_1 = 0 - 0.1 \cdot (-6) = 0 + 0.6 = 0.6$.
- Objective: $f(w_1) = (0.6 - 3)^2 = 5.76$.

Iteration 2:

- Gradient: $\nabla f(w_1) = 2(0.6 - 3) = -4.8$.

- Update: $w_2 = 0.6 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08$.
- Objective: $f(w_2) = (1.08 - 3)^2 = 3.6864$.

Iteration 3:

- Gradient: $\nabla f(w_2) = 2(1.08 - 3) = -3.84$.
- Update: $w_3 = 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464$.
- Objective: $f(w_3) = (1.464 - 3)^2 = 2.3593$.

Summary:

Iteration	w_t	$\nabla f(w_t)$	$f(w_t)$
0	0.000	-6.000	9.000
1	0.600	-4.800	5.760
2	1.080	-3.840	3.686
3	1.464	-3.072	2.359

Note: The algorithm reduces $f(w)$ iteratively but has not yet converged within 3 iterations. The gradient magnitude decreases from 6 to 3.072, remaining above $\varepsilon = 0.01$.

3 Convergence Theory

Assumptions

We formalize the assumptions required for our convergence analysis:

1. **Coordinate-Wise Smoothness:** For each coordinate $i \in \{1, 2, \dots, n\}$, the partial derivative $\nabla_i f(x)$ is Lipschitz continuous with constant $L_i > 0$. That is, for all $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$f(x + te_i) \leq f(x) + t\nabla_i f(x) + \frac{L_i}{2}t^2,$$

where e_i is the standard basis vector. Define $L_{\max} = \max_i L_i$.

2. **Polyak-Łojasiewicz (PL) Condition:** There exists $\mu > 0$ such that for all $x \in \mathbb{R}^n$,

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*),$$

where $f^* = \min_{x \in \mathbb{R}^n} f(x)$ is finite and attained.

3. **Random Coordinate Selection:** At each iteration, a coordinate $i \in \{1, 2, \dots, n\}$ is selected uniformly at random.

Main Convergence Theorem

We analyze the iterates $\{x_k\}$ generated by the randomized coordinate descent algorithm:

$$x_{k+1} = x_k - \frac{1}{L_{i_k}} \nabla_{i_k} f(x_k) e_{i_k},$$

where i_k is the coordinate selected at iteration k . The following theorem guarantees linear convergence in expectation.

[Linear Convergence under PL Condition] Let f satisfy coordinate-wise smoothness and the PL condition with parameters $\{L_i\}$ and μ , respectively. Define $\kappa = \frac{\mu}{nL_{\max}}$. Then the iterates satisfy:

$$\mathbb{E}[f(x_k) - f^*] \leq (1 - \kappa)^k (f(x_0) - f^*).$$

By coordinate-wise smoothness, for any coordinate i , we have the descent guarantee:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L_i} (\nabla_i f(x_k))^2.$$

Taking expectation over i_k , we obtain:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2n} \sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(x_k))^2.$$

Using $\sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(x_k))^2 \geq \frac{1}{L_{\max}} \|\nabla f(x_k)\|^2$, we derive:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2nL_{\max}} \|\nabla f(x_k)\|^2.$$

Applying the PL condition $\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f^*)$, we conclude:

$$\mathbb{E}[f(x_{k+1}) - f^* \mid x_k] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) (f(x_k) - f^*).$$

Recursive application completes the proof.

Theoretical Properties

Stability Analysis

The expected exponential decay of $\mathbb{E}[f(x_k) - f^*]$ ensures stability: the algorithm converges to f^* from any initial point. Convergence is not perturbed by random coordinate selection, and the iterates do not diverge even with varying L_i .

Sensitivity to Hyperparameters

The convergence rate depends critically on $L_{\max}$ and n . Larger $L_{\max}$ (e.g., ill-conditioned coordinates) or larger n slow convergence. The algorithm is robust to overestimating L_i , as smaller step sizes $1/L_i$ still satisfy the descent lemma.

Comparison to Gradient Descent

Gradient descent (GD) under PL has rate $1 - \mu/L$, where L is the global smoothness constant. Here, $L_{\max} \leq L$, so GD's rate per iteration is better by a factor of n . However, each GD iteration costs $O(n)$ times more than coordinate descent, making coordinate descent preferable in high-dimensional regimes where n is large.

Discussion

Our analysis shows that randomized coordinate descent converges linearly for functions satisfying the PL condition and coordinate-wise smoothness. The rate depends on the interplay between the PL constant μ , the maximum coordinate smoothness $L_{\max}$, and the dimension n . This extends classical results for strongly convex functions to broader non-convex settings.

Key implications include:

- **Non-Convex Compatibility:** The PL condition allows convergence guarantees for non-convex functions with sharp minima.
- **High-Dimensional Efficiency:** Coordinate descent outperforms GD in terms of total computational cost for large n , despite a slower per-iteration rate.
- **Step Size Adaptivity:** The use of coordinate-specific step sizes $1/L_i$ ensures optimal progress along each axis without prior knowledge of f^* .

Future work could explore importance sampling to mitigate dependence on $L_{\max}$, or adaptive strategies for unknown L_i . This analysis solidifies coordinate descent as a scalable method for high-dimensional optimization under weak convexity assumptions.

4 Convergence Proofs

Preliminaries (Definitions and Lemmas)

Coordinate-Wise Smoothness

For each coordinate $i \in \{1, 2, \dots, n\}$, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies coordinate-wise smoothness with constant $L_i > 0$. This implies:

$$f(x + te_i) \leq f(x) + t\nabla_i f(x) + \frac{L_i}{2}t^2 \quad \forall x \in \mathbb{R}^n, t \in \mathbb{R},$$

where e_i is the standard basis vector with 1 in the i -th coordinate and 0 elsewhere. Define $L_{\max} = \max_i L_i$.

Polyak-Łojasiewicz (PL) Condition

The function f satisfies the PL condition with parameter $\mu > 0$ if:

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x \in \mathbb{R}^n,$$

where $f^* = \min_{x \in \mathbb{R}^n} f(x)$ is finite and attained.

Random Coordinate Selection

At each iteration k , a coordinate $i_k \in \{1, 2, \dots, n\}$ is selected uniformly at random, i.e., $\mathbb{P}(i_k = i) = \frac{1}{n}$ for all i .

Key Lemmas

[Expected Decrease] For the randomized coordinate descent update:

$$x_{k+1} = x_k - \frac{1}{L_{i_k}} \nabla_{i_k} f(x_k) e_{i_k},$$

the expected function value satisfies:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2n} \sum_{i=1}^n \frac{(\nabla_i f(x_k))^2}{L_i}.$$

By coordinate-wise smoothness, for any coordinate i , setting $t = -\frac{\nabla_i f(x_k)}{L_i}$ gives:

$$f(x_{k+1}) \leq f(x_k) - \frac{(\nabla_i f(x_k))^2}{2L_i}.$$

Taking expectation over $i_k \sim \text{Uniform}(\{1, \dots, n\})$:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2n} \sum_{i=1}^n \frac{(\nabla_i f(x_k))^2}{L_i}.$$

[Gradient Norm Bound] Let $L_{\max} = \max_i L_i$ Then :

$$\sum_{i=1}^n \frac{(\nabla_i f(x))^2}{L_i} \geq \frac{1}{L_{\max}} \|\nabla f(x)\|^2. \quad (1)$$

Since $L_i \leq L_{\max}$ for all i , we have $\frac{1}{L_i} \geq \frac{1}{L_{\max}}$. Thus:

$$\sum_{i=1}^n \frac{(\nabla_i f(x))^2}{L_i} \geq \sum_{i=1}^n \frac{(\nabla_i f(x))^2}{L_{\max}} = \frac{1}{L_{\max}} \|\nabla f(x)\|^2.$$

We analyze the iterates $\{x_k\}$ generated by the randomized coordinate descent algorithm:

$$x_{k+1} = x_k - \frac{1}{L_{i_k}} \nabla_{i_k} f(x_k) e_{i_k}.$$

Step 1: Apply Lemma 1 (Expected Decrease) From Lemma 1:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2n} \sum_{i=1}^n \frac{(\nabla_i f(x_k))^2}{L_i}.$$

Step 2: Apply Lemma 2 (Gradient Norm Bound) Substitute Lemma 2 into the inequality:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{1}{2nL_{\max}} \|\nabla f(x_k)\|^2.$$

Step 3: Apply PL Condition Using the PL condition $\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f^*)$:

$$\mathbb{E}[f(x_{k+1}) \mid x_k] \leq f(x_k) - \frac{\mu}{nL_{\max}} (f(x_k) - f^*).$$

Step 4: Rearranging for Expectation Subtract f^* from both sides:

$$\mathbb{E}[f(x_{k+1}) - f^* \mid x_k] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) (f(x_k) - f^*).$$

Step 5: Take Total Expectation Taking expectation over x_k :

$$\mathbb{E}[f(x_{k+1}) - f^*] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) \mathbb{E}[f(x_k) - f^*].$$

Step 6: Recursion Unrolling the recurrence for k iterations:

$$\mathbb{E}[f(x_k) - f^*] \leq \left(1 - \frac{\mu}{nL_{\max}}\right)^k (f(x_0) - f^*).$$

Rate Derivation (With Constants)

Define $\kappa = \frac{\mu}{nL_{\max}}$. Since $0 < \kappa < 1$, we use the inequality $1 - \kappa \leq e^{-\kappa}$ to derive:

$$\mathbb{E}[f(x_k) - f^*] \leq e^{-\kappa k} (f(x_0) - f^*).$$

Thus, to achieve $\mathbb{E}[f(x_k) - f^*] \leq \epsilon$, it suffices to have:

$$k \geq \frac{nL_{\max}}{\mu} \log \left(\frac{f(x_0) - f^*}{\epsilon} \right).$$

The convergence rate is linear with per-iteration contraction factor $1 - \kappa$, and the total computational cost scales as $O(n \log(1/\epsilon))$, making it efficient for high-dimensional problems.

Special Cases

Uniform Coordinate Smoothness

If $L_i = L$ for all i , then $L_{\max} = L$, and the convergence rate becomes:

$$\mathbb{E}[f(x_k) - f^*] \leq \left(1 - \frac{\mu}{nL}\right)^k (f(x_0) - f^*).$$

Smaller L (smoother coordinates) improves the rate.

Comparison to Gradient Descent

Gradient descent under PL has rate $1 - \frac{\mu}{L}$, where L is the global smoothness constant. Since $L_{\max} \leq L$, coordinate descent has a slower per-iteration rate by a factor of n . However, each gradient descent iteration costs $O(n)$ times more than coordinate descent, making the latter preferable for large n .

Convex Functions

If f is convex and μ -strongly convex, the PL condition holds with μ equal to the strong convexity parameter. The analysis extends to this case, but does not require convexity in general.

Non-Convex Functions

The PL condition allows convergence guarantees for non-convex functions satisfying $\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*)$, which is weaker than convexity. This enables linear convergence even in non-convex settings.

Conclusion

The randomized coordinate descent algorithm converges linearly in expectation under coordinate-wise smoothness and the PL condition. The rate depends on the interplay between the PL parameter μ , the maximum coordinate smoothness $L_{\max}$, and the problem dimension n . The analysis rigorously derives all inequalities from first principles and confirms the efficiency of coordinate descent in high-dimensional regimes.